Ch 9.1: Maximum Margin Classifier

Lecture 25 - CMSE 381

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::
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Weds, Nov 16, 2022

Announcements

Last time:

- Ch 8: Random Forests

This lecture:

- Maximal Margin Classifier
- No jupyter notebook for this class

Announcements:

- HW #8 Due Friday
- No class (virtual OH only) Weds, Nov 23

20	F	Nov 4	Polynomial & Step Functions.	7.1,7.2	
21	M	Nov 7	Step Functions	7.2	
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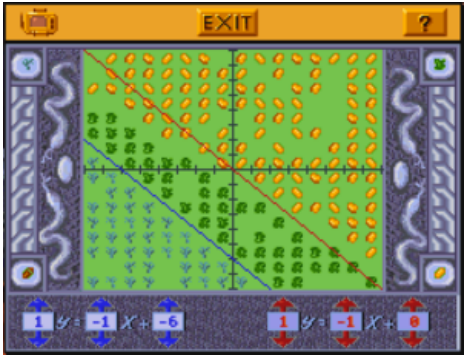
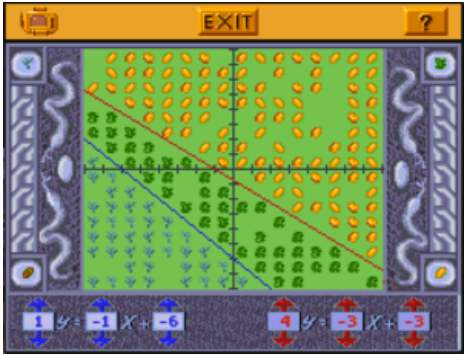
Section 1

Maximal Margin Classifier

Support Vector Machines

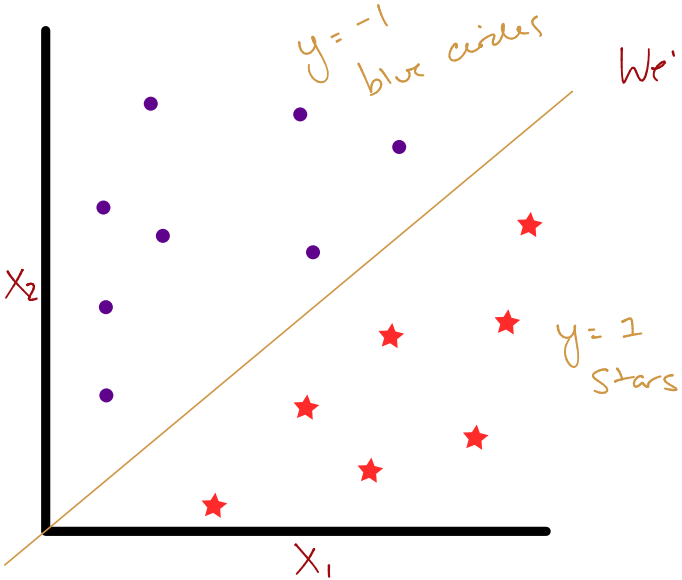
The Island of Dr Brain





https://classicreload.com/island_of_dr_brain.html

The goal



We're doing classification

2-class

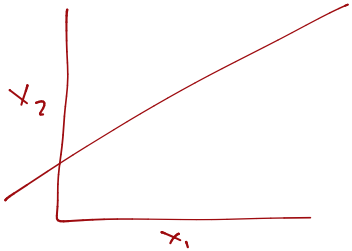
$y \in \{-1, 1\}$

What is a hyperplane?

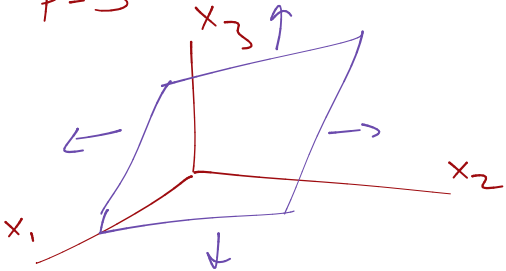
In p-dimension space $X_1, \ldots, X_p$

A hyperplane is a flat affine subspace of dimension p-1

$p = 2$



$p = 3$



Mathematical definition of a hyperplane

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p = 0$$

When $p = 2$

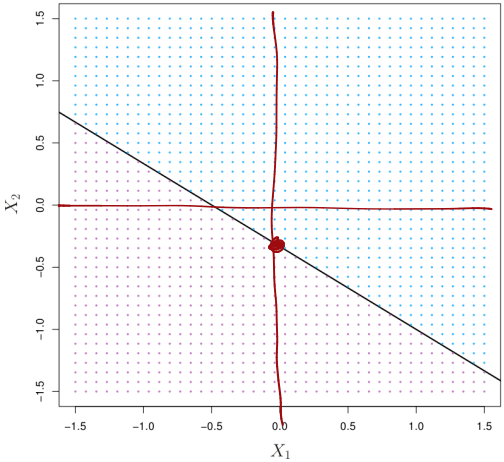
$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0$$

$$\beta_2 X_2 = -\beta_0 - \beta_1 X_1$$

$$X_2 = -\frac{\beta_0}{\beta_2} - \frac{\beta_1}{\beta_2} X_1$$

$$y = + b + m x$$

Hyperplane for $p = 2$



$$1 + 2X_1 + 3X_2 = 0$$

$$1 + 2x_1 + 3x_2 = 0$$

$$1 + 2x_1 = -3x_2$$

$$x_2 = -\frac{1}{3} - \frac{2}{3}x_1$$

There are two sides to every hyperplane

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p < 0 \qquad \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p > 0$$

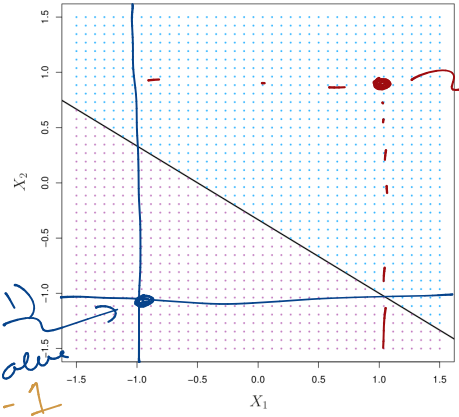
$1 + 2X_1 + 3X_2 = 0$

$1 + 2(1) + 3(1)$

$= 6$

$1 + 2(-1) + 3(-1)$

$= -4$



$(1,1)$

Positive value

$+1$

$(-1,-1)$

neg value

-1

Classification Setup

Data matrix:

$$X = \begin{pmatrix} - & x_1^T & - \\ - & x_2^T & - \\ & \vdots & \\ - & x_n^T & - \end{pmatrix}_{n \times p}$$

Observations in one of two classes, $y_i \in \{-1, 1\}$

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

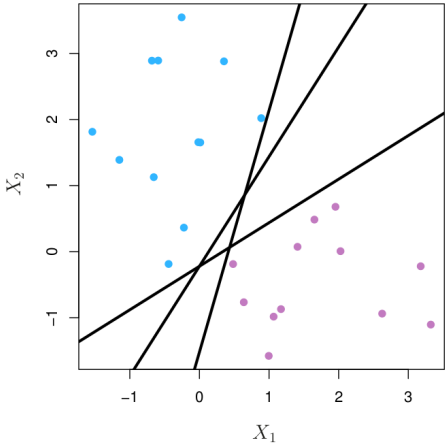
$$x_1 = \begin{pmatrix} x_{11} \\ \vdots \\ x_{1p} \end{pmatrix}, \cdots, x_n = \begin{pmatrix} x_{n1} \\ \vdots \\ x_{np} \end{pmatrix}$$

Separate out a test observation

$$x^* = (x_1^* \ \cdots \ x_p^*)^T$$

Separating Hyperplane

$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} > 0 \text{ if } y_i = 1$$
$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} < 0 \text{ if } y_i = -1$$



- Strict requirement

  No point can be on the "wrong" side

Another way to say it

$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} > 0 \text{ if } y_i = 1$$
$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} < 0 \text{ if } y_i = -1$$

For all i:

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) > 0$$

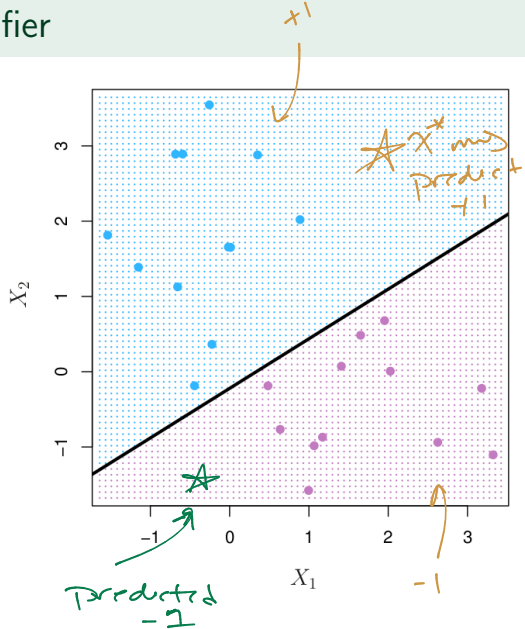
Separating hyperplane becomes a classifier

If you have a separating hyperplane:

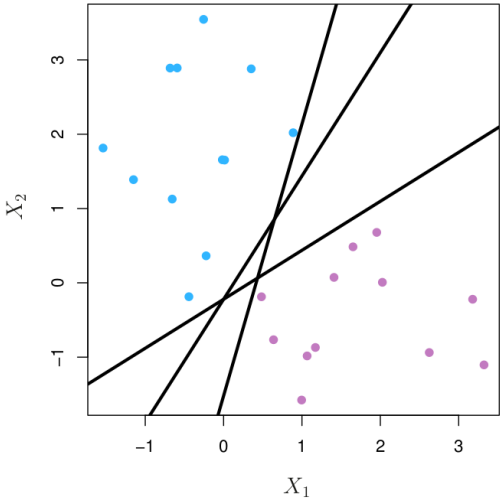
- Check

$$f(x^*) = \underbrace{\beta_0 + \beta_1 x_1^* + \beta_2 x_2^* + \cdots + \beta_p x_p^*}$$

- If positive, assign $\hat{y} = 1$
- If negative, assign $\hat{y} = -1$



How do we pick?

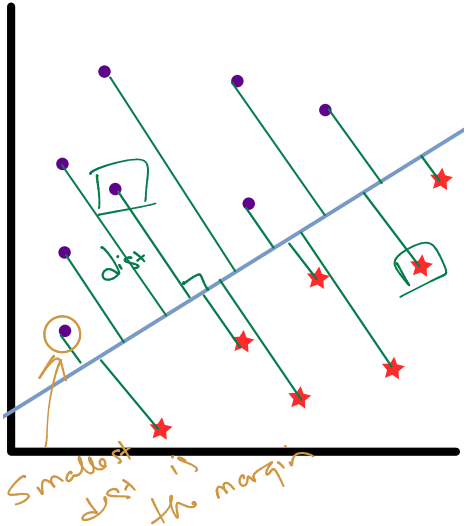


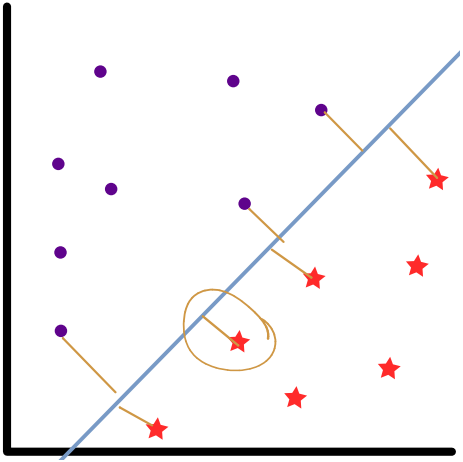
We want the "best" Seperating hyperplane

For each data point, compute distance from the point to the hyperplane

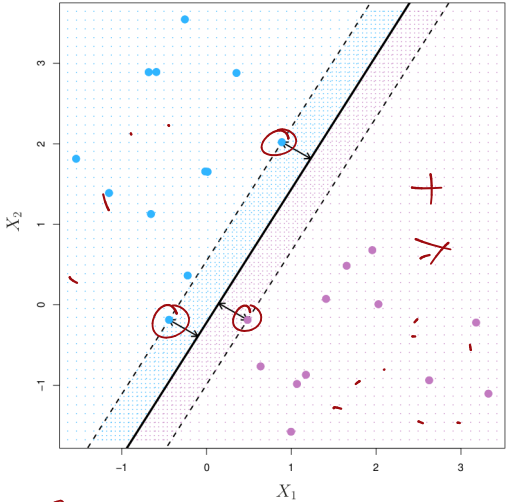
The smallest such distance is called the <u>margin</u>

Distance from an observation to a hyperplane





Maximal margin classifier



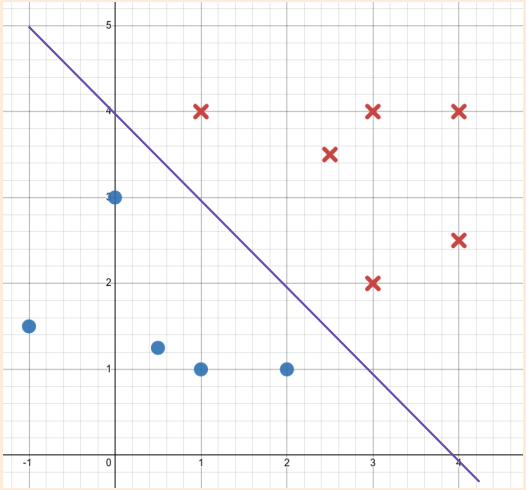
Maximal margin hyperplane is the seperating hyperplane with the largest margin

The classifier using this hyper plane is the maximal margin hyperplane.

Observations that are closest to the hyperplane are called support vectors

These are the points that "decide" the best hyperplane.

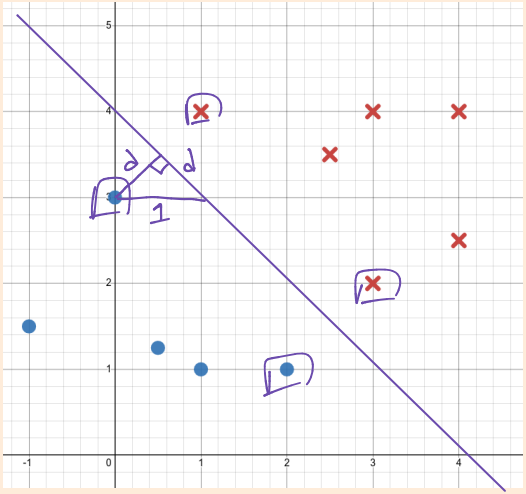
Example



- Sketch the maximal margin hyperplane.
- What is the equation of this line in the form $\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0$?
- Circle the support vectors. What is their distance from the line?

desmos.com/calculator/lqms253gfq

Extra work space



$$y = -1x + 4$$

$$x_2 = -x_1 + 4$$

$$x_1 + x_2 - 4 = 0$$

$$d^2 + d^2 = 1$$

$$2d^2 = 1$$

$$d^2 = \frac{1}{2}$$

$$d = \frac{1}{\sqrt{2}}$$

desmos.com/calculator/lqms253gfq

Mathematical Formulation

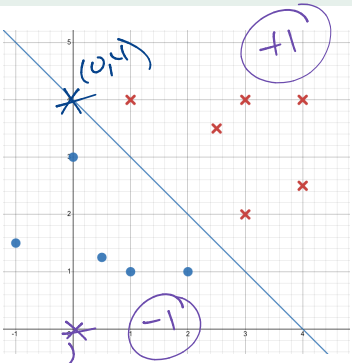
$$\underset{\beta_0,\beta_1,\ldots,\beta_p,M}{\text{maximize}} \quad M$$

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$\underbrace{y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip})}_{\text{distance to the hyperplane}} \geq \underbrace{M}_{M = \text{minimum dist to the hyperplane}} \quad \forall\, i = 1, \ldots, n$$

(because $\sum_{i=1}^{p} \beta_i^2 = 1$)

First constraint



$$\sum_{i=1}^{p} \beta_i^2 = 1$$

$$0 = \beta_0 + \frac{1}{\sqrt{2}} x_1 + \frac{1}{\sqrt{2}} x_2$$

$$\boxed{\underbrace{-4}_{\beta_0} + \underset{\beta_1 = 1}{x_1} + \underset{\beta_2 = 1}{x_2}}$$

$$0 = \beta_0 + 0 + \frac{1}{\sqrt{2}} 4$$

$$\beta_0 = -\frac{4}{\sqrt{2}}$$

$$1^2 + 1^2 = 2 \neq 1$$

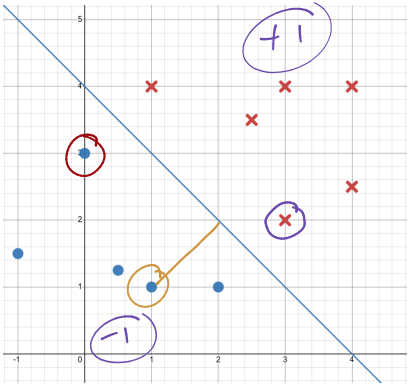
$$\boxed{-\frac{4}{\sqrt{2}} + \frac{1}{\sqrt{2}} x_1 + \frac{1}{\sqrt{2}} x_2}$$

$f(x_1, x_2)$

Divide coeff by $\left(\sum \beta_i^2\right)^{1/2}$, can get a version satisfying constrains

$$f(0,0) = -\frac{4}{\sqrt{2}} + 0 + 0 < 0$$

Second constraint



$$\overbrace{y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2})}^{\text{dist to the hyperplane}} \geq M$$

pt $(3,2)$, $y_i = +1$

$$-2\sqrt{2} + \frac{\sqrt{2}}{2}3 + \frac{\sqrt{2}}{2}\cdot 2 = \frac{\sqrt{2}}{2}$$

$$y_i\left(\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}} = \text{distance to the hyperplane}$$

pt $(0,3)$ $y_i = -1$

$$-2\sqrt{2} + 0 + \frac{\sqrt{2}}{2}(3) = -\frac{\sqrt{2}}{2}$$

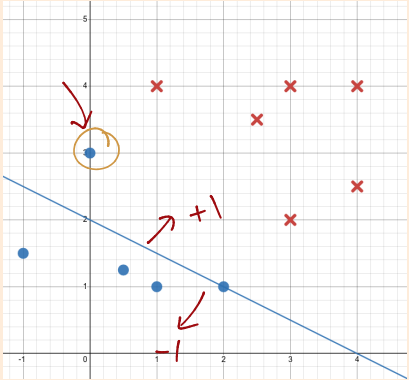
$$y_i\left(-\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{2}}{2}$$

pt $(1,1)$ $y_i = -1$

$$-2\sqrt{2} - \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = -\sqrt{2}$$

- Blue circles: $y_i = -1$
- Red Xs: $y_i = 1$
- $-2\sqrt{2} + \frac{\sqrt{2}}{2}X_1 + \frac{\sqrt{2}}{2}X_1 = 0$

An example with a bad choice of hyperplane



What is $y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2})$ for the point $x_i = (0, 3)$?

$y_i = -1$

$$-1\left(-\frac{4}{\sqrt{5}} + 0 + \frac{2}{\sqrt{5}} \cdot 3\right) = \frac{-2}{\sqrt{5}}$$

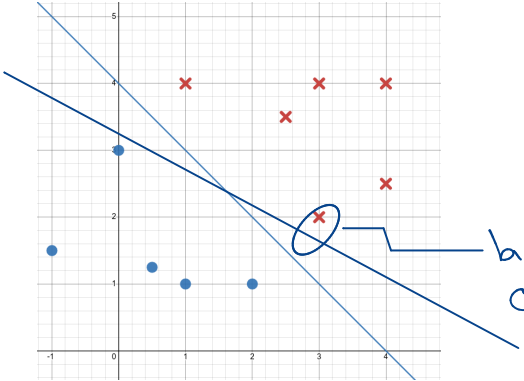
Not a seperating hyperplane

- Blue circles: $y_i = -1$
- Red Xs: $y_i = 1$
- $-\frac{4}{\sqrt{5}} + \frac{1}{\sqrt{5}} X_1 + \frac{2}{\sqrt{5}} X_2 = 0$

Second constraint extra space

Maximize M

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}) \geq M$$



biggest M is the closest pt distance

- Blue circles: $y_i = -1$
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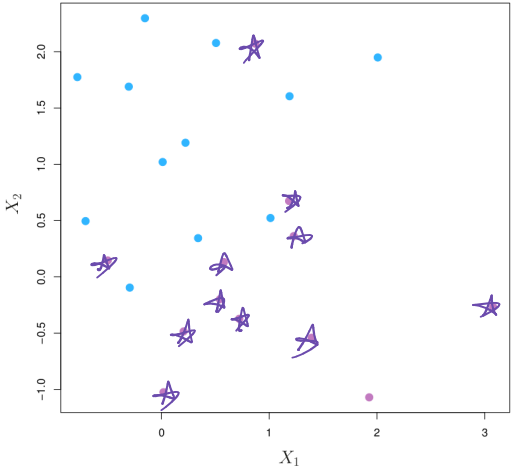
Section 2

Issues with Maximal Margin Classifier

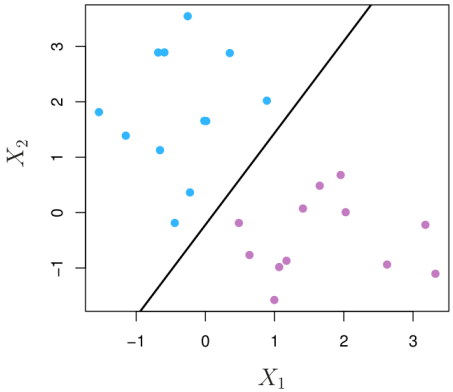
But what if....

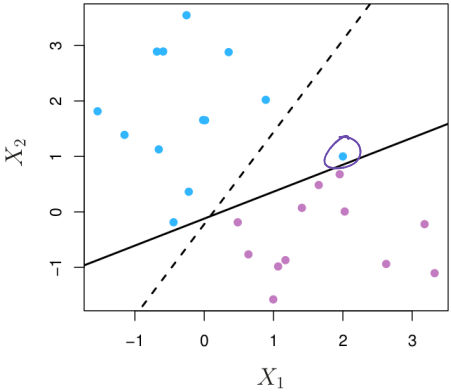


There is no separating hyperplane?

Next time...
the fix

Sensitivity to new points





Addition of one point moves the hyperplane a lot!

Next time

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